CHAPTER 4

Polynomial equations

Quadratics in $\mathbb{C}$, factorisation, number of roots

Objectives. Solve in $\mathbb{C}$ a quadratic equation with real coefficients; factorise a polynomial by $z-a$ when a is a root; solve a cubic equation when one root is known; know that a polynomial of degree n has at most n roots.

Prerequisites. Quadratics in $\mathbb{R}$ (completed square form, discriminant, sum and product of the roots); chapters 1 and 2; sum of the terms of a geometric sequence.

Fluency

Fluency · No calculator, five minutes.

1. Compute the discriminant of $2x^2-3x+5$.
2. Factorise x^2-5x+6.
3. Expand $(x-2)\left(x^2+2x+4\right)$.
4. Compute $P(2)$ for $P(x)=x^3-3x^2+4$.
5. Write $1+q+q^2+\cdots+q^n$ as a quotient, for $q\neq 1$.
6. Give the sum and the product of the roots of $x^2-7x+10$.

Solution · Fluency.

1. $\Delta = 9-40=-31$.
2. $(x-2)(x-3)$.
3. $x^3+2x^2+4x-2x^2-4x-8=x^3-8$.
4. $P(2)=8-12+4=0$.
5. $\dfrac{1-q^{n+1}}{1-q}$.
6. Sum 7, product 10 (roots 2 and 5).

Introductory activity

Activity 1 · One root hides two others 35 min . Consider $P(z) = z^3 - 2z^2 + 4z - 8$.

1. Plot on a calculator the curve of $x \mapsto x^3 - 2x^2 + 4x - 8$ on $[-3, 4]$. How many real solutions does the equation $P(x) = 0$ seem to have? Locate one.
2. Check that $P(2) = 0$.
3. Find real numbers a, b, c such that, for every z, $P(z) = (z-2)\left(az^2 + bz + c\right)$. Expand the right-hand side and identify the coefficients.
4. Deduce all the complex solutions of $P(z) = 0$.
5. Compute the sum and the product of the three solutions. Compare with the coefficients of P.
6. How many complex solutions can a polynomial of degree 3 have at most? State a conjecture for degree n.

Observing is not proving: the properties conjectured here are established in the course.

Solution · Activity.

1. The curve crosses the horizontal axis only once, at $x = 2$: a single real solution.
2. $8 - 8 + 8 - 8 = 0$.
3. $(z-2)\left(az^2 + bz + c\right) = az^3 + (b-2a)z^2 + (c-2b)z - 2c$. Identification gives $a = 1$, $b - 2a = -2$, $c - 2b = 4$, $-2c = -8$, that is $a = 1$, $b = 0$, $c = 4$ (the last equation is consistent). So $P(z) = (z-2)\left(z^2 + 4\right)$.
4. $z^2 + 4 = (z - 2i)(z + 2i)$: the solutions are 2, $2i$ and $-2i$.
5. Sum: 2; product: $2 \times 2i \times (-2i) = 8$. For $z^3 + az^2 + bz + c$, we observe sum $= -a$ and product $= -c$.
6. At most three. Conjecture: at most n for degree n.

Teacher note · Running the activity. Identifying coefficients relies on a property that the syllabus takes for granted: a zero polynomial function has all its coefficients equal to zero. Point it out explicitly. Question 5 prepares the problem (relations between coefficients and roots).

Course notes

Quadratics with real coefficients

Property · Solving in $\mathbb{C}$. Let a, b, c be three real numbers with $a \neq 0$, and $\Delta = b^2 - 4ac$. The equation $az^2 + bz + c = 0$ has:

1. if $\Delta > 0$, two real solutions $\dfrac{-b - \sqrt{\Delta}}{2a}$ and $\dfrac{-b + \sqrt{\Delta}}{2a}$;
2. if $\Delta = 0$, one real solution $-\dfrac{b}{2a}$;
3. if $\Delta < 0$, two conjugate complex solutions $z_1 = \dfrac{-b - i\sqrt{-\Delta}}{2a}$ and $z_2 = \dfrac{-b + i\sqrt{-\Delta}}{2a}$.

In every case, $az^2 + bz + c = a(z - z_1)(z - z_2)$, $z_1 + z_2 = -\dfrac{b}{a}$ and $z_1 z_2 = \dfrac{c}{a}$ (with $z_1 = z_2$ if $\Delta = 0$).

Proof. The completed square form $az^2 + bz + c = a\left[\left(z + \frac{b}{2a}\right)^2 - \frac{\Delta}{4a^2}\right]$ still holds in $\mathbb{C}$. If $\Delta < 0$, write $\frac{\Delta}{4a^2} = \left(\frac{i\sqrt{-\Delta}}{2a}\right)^2$, then factorise the difference of two squares. A product of complex numbers is zero if and only if one of the factors is zero (since every nonzero complex number is invertible). Sum and product are obtained by expanding $a(z - z_1)(z - z_2)$ and identifying. □

Example 1 · Solving $z^2 - 2z + 5 = 0$. $\Delta = 4 - 20 = -16 = (4i)^2$, so $z = \frac{2 \pm 4i}{2}$: the solutions are $1 - 2i$ and $1 + 2i$.

Factorisation by $z - a$

Definition · Polynomial function, root. A *polynomial function* of degree n is a function $P : z \mapsto a_n z^n + \cdots + a_1 z + a_0$ defined on $\mathbb{C}$, where $a_0, \ldots, a_n$ are complex numbers and $a_n \neq 0$. A complex number α such that $P(\alpha) = 0$ is a *root* of P.

Key point · Property taken for granted. If a polynomial function is zero for every complex number z, then all its coefficients are zero. Consequently, two equal polynomial functions have the same coefficients: we may *identify* the coefficients.

Property · Factorisation of $z^n - a^n$. For all complex numbers z, a and every integer $n \geq 1$:

$$z^n - a^n = (z - a)\left(z^{n-1} + az^{n-2} + \cdots + a^{n-2}z + a^{n-1}\right) = (z - a)\sum_{k=0}^{n-1} a^k z^{n-1-k}.$$

Proof. Expand: $(z - a)\sum_{k=0}^{n-1} a^k z^{n-1-k} = \sum_{k=0}^{n-1} a^k z^{n-k} - \sum_{k=0}^{n-1} a^{k+1} z^{n-1-k}$. Setting $j = k + 1$ in the second sum, it becomes $\sum_{j=1}^{n} a^j z^{n-j}$. The terms with index 1 to $n - 1$ cancel; what remains is the term $k = 0$ of the first sum, z^n, and the term $j = n$ of the second, a^n. We obtain $z^n - a^n$. □

Property · Factorisation by $z - \alpha$. Let P be a polynomial function of degree $n \geq 1$ and α a root of P. There is a polynomial function Q of degree $n - 1$ such that, for every complex number z, $P(z) = (z - \alpha)Q(z)$.

Proof. Write $P(z) = \sum_{k=0}^{n} a_k z^k$. Since $P(\alpha) = 0$,

$$P(z) = P(z) - P(\alpha) = \sum_{k=1}^{n} a_k \left(z^k - \alpha^k\right).$$

By the previous theorem, each $z^k - \alpha^k$ can be written $(z-\alpha)Q_k(z)$ with Q_k a polynomial of degree $k-1$ with leading coefficient 1. So $P(z) = (z-\alpha)Q(z)$ with $Q = \sum_{k=1}^{n} a_k Q_k$, of degree $n-1$ with leading coefficient $a_n \neq 0$. □

Property · Number of roots. A polynomial function of degree $n \geq 1$ has at most n roots.

Proof. By induction on n. A polynomial function of degree 1, $a_1 z + a_0$ with $a_1 \neq 0$, has exactly one root, $-\frac{a_0}{a_1}$. Assume the property holds in degree n, and let P have degree $n+1$. If P has no root, we are done. Otherwise, let α be a root: $P(z) = (z-\alpha)Q(z)$ with Q of degree n. If $P(\beta) = 0$, then $\beta = \alpha$ or $Q(\beta) = 0$. Since Q has at most n roots, P has at most $n+1$. □

Property · Conjugate roots. If P has real coefficients and α is a root of P, then $\overline{\alpha}$ is also a root of P.

Proof. Since the coefficients are real, $P\left(\overline{\alpha}\right) = \sum a_k \overline{\alpha}^k = \overline{\sum a_k \alpha^k} = \overline{P(\alpha)} = 0$. □

Method · Cubic equation with a known root. Factorise by $z - \alpha$ by identifying the coefficients of $Q(z) = az^2 + bz + c$, then solve $Q(z) = 0$. Example: $z^3 + 8 = 0$ has the root -2 and $z^3 + 8 = (z+2)\left(z^2 - 2z + 4\right)$. The quadratic has discriminant $-12 = \left(2i\sqrt{3}\right)^2$, hence the solutions -2, $1 - i\sqrt{3}$, $1 + i\sqrt{3}$.

Watch out. The discriminant of a quadratic with real coefficients no longer tells whether the equation has solutions in $\mathbb{C}$: it always has. It tells whether they are real or conjugate complex numbers.

Exercises

Exercise 4.1 · Quadratics. Solve in $\mathbb{C}$: $z^2+9=0$; $z^2-4z+13=0$; $2z^2+2z+1=0$; $z^4-1=0$.

Solution. $z^2+9=(z-3i)(z+3i)$: $\{-3i, 3i\}$. Next $\Delta=-36$: $2-3i$ and $2+3i$. Then $\Delta=-4$: $\frac{-2\pm 2i}{4}$, that is $-\frac{1}{2}-\frac{1}{2}i$ and $-\frac{1}{2}+\frac{1}{2}i$. Finally $z^4-1=\left(z^2-1\right)\left(z^2+1\right)$: $\{1,-1,i,-i\}$.

Exercise 4.2 · Reducing to a quadratic.

1. Solve in $\mathbb{C}$ the equation $z^4+4z^2-5=0$ by setting $Z=z^2$.
2. Solve in $\mathbb{C}$ the equation $z+\dfrac{1}{z}=1$, for $z\neq 0$.

Solution.

1. $Z^2+4Z-5=(Z-1)(Z+5)$. So $z^2=1$ or $z^2=-5$: $\{1,-1,i\sqrt{5},-i\sqrt{5}\}$.
2. For $z\neq 0$, the equation is equivalent to $z^2-z+1=0$, with discriminant -3: $z=\frac{1-i\sqrt{3}}{2}$ or $z=\frac{1+i\sqrt{3}}{2}$, both nonzero.

Exercise 4.3 · An obvious root. Let $P(z)=z^3-3z^2+4z-2$.

1. Check that 1 is a root of P.
2. Determine the polynomial Q such that $P(z)=(z-1)Q(z)$.
3. Solve $P(z)=0$ in $\mathbb{C}$.

Solution.

1. $1-3+4-2=0$.
2. $(z-1)\left(z^2+bz+c\right)=z^3+(b-1)z^2+(c-b)z-c$, hence $b=-2$ and $c=2$: $Q(z)=z^2-2z+2$.
3. Q has discriminant -4: roots $1-i$ and $1+i$. The solutions are 1, $1-i$, $1+i$.

Exercise 4.4 · A non-real root. Let $P(z)=z^3+z^2+4z+4$.

1. Check that $2i$ is a root of P. Deduce another root without computation.
2. Expand $(z-2i)(z+2i)$, then factorise $P(z)$ and solve $P(z)=0$.

Solution.

1. $P(2i)=-8i-4+8i+4=0$. The coefficients are real, so $-2i=\overline{2i}$ is also a root.
2. $(z-2i)(z+2i)=z^2+4$ and $P(z)=z^2(z+1)+4(z+1)=(z+1)\left(z^2+4\right)$. The solutions are -1, $2i$, $-2i$.

Exercise 4.5 · Factorising $z^3 - 27$.

1. Factorise $z^3 - 27$ by $z - 3$ using the course.
2. Solve in $\mathbb{C}$ the equation $z^3 = 27$ and plot the solutions in the complex plane. What do you notice?

Solution.

1. $z^3 - 3^3 = (z-3)\left(z^2 + 3z + 9\right)$.
2. The quadratic has discriminant $9 - 36 = -27 = \left(3i\sqrt{3}\right)^2$: $z = \frac{-3 \pm 3i\sqrt{3}}{2}$. The three solutions 3, $\frac{-3-3i\sqrt{3}}{2}$, $\frac{-3+3i\sqrt{3}}{2}$ have modulus 3 and form an equilateral triangle inscribed in the circle with centre O and radius 3 (chapter 5).

Exercise 4.6 · A parameter. Let m be a real number. Consider the equation (E_m): $z^2 - 2z + m = 0$.

1. Depending on the value of m, give the number and the nature of the solutions of (E_m).
2. Assume $m > 1$. Let A and B be the images of the solutions. Determine m so that the triangle OAB is equilateral.

Solution.

1. $\Delta = 4 - 4m$. If $m < 1$: two real solutions $1 \pm \sqrt{1-m}$; if $m = 1$: one solution, 1; if $m > 1$: two conjugate solutions $1 \pm i\sqrt{m-1}$.
2. $OA = OB = \sqrt{1 + (m-1)} = \sqrt{m}$ and $AB = 2\sqrt{m-1}$. The triangle is equilateral if and only if $m = 4(m-1)$, that is $m = \frac{4}{3}$ (which satisfies $m > 1$).

Exercise 4.7 · Recovering a polynomial. Determine the polynomial P of degree 3, with real coefficients and leading coefficient 1, such that $1 + i$ is a root of P and $P(0) = 4$.

Solution. Since the coefficients are real, $1 - i$ is also a root. Factorise P by $z - (1+i)$; the quotient, of degree 2, vanishes at $1 - i$ (because $1 - i \neq 1 + i$), so it can in turn be factorised by $z - (1-i)$. What remains is a factor of degree 1 with leading coefficient 1: $P(z) = (z-1-i)(z-1+i)(z-r) = \left(z^2 - 2z + 2\right)(z-r)$ for some complex number r. $P(0) = -2r = 4$, so $r = -2$ and $P(z) = \left(z^2 - 2z + 2\right)(z+2) = z^3 - 2z + 4$. Check: $P(0) = 4$ and $P(1+i) = (-2+2i) - 2 - 2i + 4 = 0$.

Problem

Problem · Coefficients and roots of a cubic polynomial. Let $P(z) = z^3 + az^2 + bz + c$ be a polynomial with complex coefficients, with three roots z_1, z_2, z_3 (not necessarily distinct), so that $P(z) = (z - z_1)(z - z_2)(z - z_3)$ for every complex number z.

1. By expanding, prove that
$$z_1 + z_2 + z_3 = -a, \qquad z_1z_2 + z_1z_3 + z_2z_3 = b, \qquad z_1z_2z_3 = -c.$$
2. Check these relations on the polynomial of the activity, $z^3 - 2z^2 + 4z - 8$.
3. Prove that $z_1^2 + z_2^2 + z_3^2 = a^2 - 2b$.
4. Consider $R(z) = z^3 - 2z^2 + 3z - 5$, which we take for granted can be written $(z - z_1)(z - z_2)(z - z_3)$. Compute $z_1^2 + z_2^2 + z_3^2$ without solving the equation.
5. Deduce that the roots of R are not all real.

Beyond the syllabus · The d'Alembert-Gauss theorem. Every polynomial of degree $n \geq 1$ with complex coefficients factorises in $\mathbb{C}$ as a product of n factors of degree 1. This theorem is not part of the syllabus: it is admitted in question 4 and is not assessed.

Solution · Problem.

1. $(z - z_1)(z - z_2)(z - z_3) = z^3 - (z_1 + z_2 + z_3)z^2 + (z_1z_2 + z_1z_3 + z_2z_3)z - z_1z_2z_3$. Identifying the coefficients (property taken for granted) gives the three relations.
2. The roots are 2, $2i$, $-2i$: sum $2 = -(-2)$; sum of the pairwise products $4i - 4i + 4 = 4$; product $8 = -(-8)$.
3. $(z_1 + z_2 + z_3)^2 = z_1^2 + z_2^2 + z_3^2 + 2(z_1z_2 + z_1z_3 + z_2z_3)$, so $z_1^2 + z_2^2 + z_3^2 = a^2 - 2b$.
4. $a = -2$ and $b = 3$: $z_1^2 + z_2^2 + z_3^2 = 4 - 6 = -2$.
5. If the three roots were real, the sum of their squares would be nonnegative. It equals -2: at least one root is not real.

Teacher note · Status of the problem. Vieta's formulas are a "possible problem" of the syllabus. The fact that every polynomial of degree 3 factorises in this way over $\mathbb{C}$ (the d'Alembert-Gauss theorem) is beyond the syllabus: it is taken for granted in the statement. For R, the associated real function is strictly increasing (its derivative $3x^2 - 4x + 3$ has a negative discriminant): one real root, two non-real conjugate roots.